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EPC Infrastructure & Power Generation | 25+ Years

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When commercial software doesn't solve the problem, I build the solution and deploy it in production.

146	13	5	\$1.6B+	25+
Indexed documents	Portfolio series	Proprietary systems	EPC value	Years

Portfolio Catalog | Civil Engineering & Geospatial Solutions | July 2026

EXECUTIVE SUMMARY

Portfolio at a Glance

This catalog documents 25 years of technical work on large-scale EPC infrastructure — dams, TBM tunnels, underground hydropower powerhouses, mountain roads, and land development. Each of the 13 series below answers a different question: *Can this engineer execute at this scale? Can he build the tools he needs when they don't exist commercially? Can he deliver auditable results under pressure?*

Central principle: When commercial software doesn't solve the problem, build the solution and deploy it in production. The same logic that produced SALT in 2004 for cadastral surveys produced ICOD in 2020 to process 2,800 LiDAR sections in 2 weeks. The physical object changes. The thinking architecture doesn't.

Proprietary Systems — Built & Deployed in Production

System	Origin	Key metric	Status
AC&P	2004	80 APU · \$508K NC demo · inverse engine 1.101...	Active — US v503A Apr 2026
Visual CRP	2007	15,000+ records · 60+ users · \$1.6M+ savings · 9 firms	Live — PMG autonomous since 2021
SALT	2004	Method PB · 1:516,577 · 32,390m · residual avg 0.44"	Rev 4.0 · public demo 85MB
ETL Links	2008	ExecuteExcel4Macro · 253 fields · N files unlimited	Rev 5.0 · Series 517A
ICOD	2020	2,800 sections · 4,000+ pts/section · 75% drafting reduction	Exclusive to Montegrande

Series Overview — 13 Series, 146 Documents

Series	Title	Key differentiator
000-014	Major Projects Overview	PMG \$500M+ · Palomino \$400M+ · TBM 12.44km · Las Placetas +27%
101-105	Tunnel Laser Scanner (ICOD)	2,800 auto sections · 4,000+ pts/section · 'Bajar 0.2000m' diagnostic
201-207	GIS & Geomatic Applications	4,110,752 m3 certified · BIM-GIS · GPS/GNSS 5 stations 12h
208-223	Surveying Processing Pipeline	UAV DJI→Pix4D→CAD · 124-sheet Doc 212 · SALT 1:52,753/16.2km
300-309	Roadway Design & Hydraulics	Rational Method 5 basins · 72" culvert · 18 AASHTO 1993 alternatives
400-416	3D Models · As-Built · Land Dev	Cavern 47.35x33m · Factor Scale 1.0000745 · 390 units Google Earth
500-517A	Excel VBA Applications	5 proprietary systems · inverse engine · Method PB · ETL 253 fields
502	Visual CRP User Training	9 firms · tripartite flow · 19 years competitive analysis
600-616E	Engineering Reports & Case Studies	+18% area → +36% concrete (quadratic) · EGM2008 · CERCONS validation · SNIP 17248 Infrastructure Audit USD 932.8M
700-707	Quantity Take-off	\$708K certified · 928K m3 cut + 1.8M m3 fill · \$500M+ impasse resolved
800-807	Estimations & Drafting Svc	4 CAASD drawings 48h · 7-field coding · ITRF-2000 · dual signature
901-916	Excavation Advance	16 periods drone · 3.74M m3 · Sector C 38% advance / 69% deficit

Online access: <https://visualcrp.com/portfolio.html>

SERIES 000-014

Major Projects Overview

Montegrande Dam \$500M+ | Palomino Hydropower \$400M+ | Las Placetas \$700M+ | TBM 12.44km

\$900M+	12.44km	15	+27%
Combined EPC value	Double-shield TBM	Visual documents	Las Placetas gain

This series establishes project scale before any technical detail is reviewed. Fifteen visual documents across three megaprojects provide irrefutable context: the Montegrande reservoir is verifiable on Google Earth (26 Jan 2024); TBM launch-sequence frames show the machine before it disappeared 12.44km underground; Las Placetas documents a paralyzed \$700M+ project reactivated and redesigned for +27% generation capacity in six months.

Montegrande Dam (PMG) — Docs 000-002

Doc 000	PMG — Aerial drone, completed dam	Drone zenith view. Reservoir full on Rio Yaque del Norte. L=2.2km H=57m crown visible. Physical evidence of \$500M+ constructed — not a render.
Docs 001-002	Centro Poblado — 390 homes	Construction → completed sequence. 390 units in elevated zone above flood area. All affected communities relocated. Verifiable Google Earth 26/01/2024.

Palomino Hydropower — Docs 003-014 (TBM chronological sequence)

Doc 003	Panoramic annotated — powerhouse site	Helicopter view, 20+ components annotated in cyan: TBM Platform, Headrace Portal, Doval Factory, Helipad, CCR Dam. Author has this site memorized.
Doc 006	TBM in position — highest technical value	Cutterhead red/blue visible. Single access road: any failure immobilizes a \$40M+ machine. Surveying is not support — it is the enabler.
Docs 008-011	TBM launch sequence	Visual timeline of the start of the 12.44km drive. Belt conveyor curving to surface. Most engineers who 'worked with TBM' first see the machine already inside. These frames show the moment before 12.44km underground.

Las Placetas — Project paralyzed since 2009. Promoted to Interim Engineering Manager. In 6 months: Alternative Basic Design approved. Jagua 82.92MW + Bao 28.40MW = 111.32MW total · 414.81 GWh/yr · +27% generation. Two RCC dams + 20.16km tunnels.

SERIES 101-105

Tunnel Laser Scanner Application — ICOD v2.0

Montegrande Dam · Diversion Tunnels Ø9.20m | EXCLUSIVE to Montegrande

2,800	4,000+	12+ km	75%
Auto cross-sections	LiDAR pts / section	Tunnels processed	Drafting reduction

No commercial tool could process LiDAR point-cloud density at this scale inside an active tunnel. ICOD v2.0 was built in two weeks: Excel VBA driving AutoCAD Civil 3D via DDE (Dynamic Data Exchange), generating 2,800 cross-sections fully automatically. Author credit explicit on screen: *ICOD versión 2.0 — Autor: Ing. Piter J. Bonilla D.*

Doc 101	MP4 — scanner in field	Scanner operating inside diversion tunnel. ~4,000 points per section in difficult access conditions. Not a lab — active construction site.
Doc 102	ICOD interface screenshot	UX with section table: North, East, Elevation, offsets, zenith angle. Author credit on screen.
Doc 103	MP4 — Civil 3D via DDE	DDE between Excel VBA and AutoCAD Civil 3D 2008. Full cross-section generated without manual point-by-point intervention.
Doc 104	Section Est. 0+138 — diagnostic	Three layers: theoretical Ø9.20m, measured, previous. Annotation: 'Bajar 0.2000 m' — actionable correction. ICOD doesn't only draw: it diagnoses.
Doc 105	3D stacked section view	2,800 sections in 3D perspective. Longitudinal variation immediately visible. What took 2–3h per section: automated.

SERIES 201-207

GIS & Geomatic Applications

Monte grande Dam + Las Placetas Hydropower

4.11M	5 stn	15 km	2
m3 certified Global Mapper	GPS/GNSS 12h sessions	LiDAR corridor HLP	Projects covered

Complete geospatial ecosystem from first-order geodetic control to 3D structural model integration with real LiDAR terrain — across two distinct megaprojects. Covers what the US market designates **BIM-GIS integration** in heavy infrastructure.

Img 1-3	Civil 3D 3D model — PMG	Three views: real LiDAR photorealistic terrain + red grid lines + structural 3D model of dam. BIM-GIS integration — differentiating capability in heavy infrastructure.
Img 4	Global Mapper Pro v23 — spillway	Volume certified: 4,110,752 m3 in a single run. WGS84 UTM Zone 19N. 8 active layers including LiDAR TIN from 4 flight campaigns. Workspace saved and auditable.
Img 6	Las Placetas — LiDAR 2007	DTM 2007 · 1:150,000 · 15–20km hydraulic corridor. Directly applicable to transmission line and infrastructure corridor desktop studies.
Img 7	Topcon Tools — GPS/GNSS network	5 simultaneous stations · 12h sessions · Geoid separations EGM2008 (characteristic negative Caribbean values) · Scale factors confirmed UTM Zone 19N. First-order geodetic control.

SERIES 208-223

Surveying Processing & Applications

SALT · RTK/GPS · Pix4D · Palomino Tunnels Geometry

1:52,753	124	21	00.44"
SALT precision · 16.2km	Sheets Doc 212	CORS stations RD	Method PB avg residual

Doc 208	Pix4D Cloud — full UAV pipeline	DJI Phantom 4 Pro v2 → Pix4D → collaborative publishing → CAD integration. End-to-end UAV pipeline operational in production.
Doc 212	Tunnels Geometry Palomino — CROQUIS-027 Rev. C	ELABORATED and VERIFIED: P. Bonilla. APPROVED: P. Kenzo (Odebrecht). Four days: Rev A → Rev B → Rev C. 124 sheets · 7 interconnected tunnels under dual logo Odebrecht+EGEHID.
Docs 215-216	RTK/GPS course — instructor level	NGS RT1-RT4 table: RT1 achieves 0.015m horizontal / 0.025m vertical. SALT as course bonus with YouTube demo. Instructor-level expertise.
Docs 218-223	SALT system — 6 documents	Rev 4.0. Method PB: 50 points · 32,390m · avg residual 00.44" · 1:516,577 precision · closure Horz=0.001m. youtube.com/watch?v=opBY0S1D1R0

SERIES 300–309

Roadway Design & Hydraulics

Palomino + Las Placetas | AASHTO 1993 | Rational Method

18			72"	5M	CBR=3%
AASHTO 1993 alternatives			Max culvert designed	W18 equivalent axles	Weak tropical subgrade
Doc 307	Powerhouse access drainage		Rational Method + Kirpich for 5 basins. I(5min)=157.47mm/hr TR=20yr. Prog 0+173.91: Q=6.43m3/s → Ø72" / 29.00m / 28 pipes H.fill=6.65m.		
Doc 309	18 pavement alternatives — AASHTO 1993		W18=5,000,000 · R=95% (ZR=-1.645) · CBR=3% · SN=4.89. Three families: granular, cement-stabilized, high-strength. Signed P. Bonilla author and reviewer.		
Docs 301–306	Geometric design — Las Placetas + Palomino		Alignment alternatives, plan-profile, cross-sections, dam access road, powerhouse road. Full EPC-standard design documentation.		

SERIES 400-416

3D Models · As-Built · Land Development

Palomino + Montegrande | Underground Powerhouse | Centro Poblado

			47.35m	99.50m	390	1.0000745
			Cavern length	Rock overburden	Units executed	Combined scale factor
Doc 406	Cavern Casa de Máquinas — Croquis-117 Rev. B	Crown EL.427.10m · Bridge crane rail EL.422.90m · Floor EL.418.85m · Turbine pit EL.415.20m · Draft tube EL.413.00m. 47.35m × 33.00m · 99.50m overburden. Signed P. Bonilla design + review.				
Doc 407	Surge shaft as-built — Rev. J (10 iterations)	Combined scale factor 1.0000745 — mathematical proof of first-order geodetic control: 7.45mm per 100m measured. Rev J = 10th edition = monthly tracking.				
Docs 409-416	Centro Poblado PMG — Land development	163,385m3 excavation · 84,227m3 fill · 51,851m2 paved streets · 5,489m sanitary sewer · 98 manholes. Cross-validated vs INAPA. Verifiable Google Earth Jan 2024.				

SERIES 500–517A

Excel VBA Applications — 5 Proprietary Systems

Visual CRP · AC&P · SALT · Section Automation · ETL Links

5	19+ yrs	\$1.6B+	253
Proprietary systems	Visual CRP production	Project value supported	ETL configurable fields

This series answers the question no resume can answer directly: can this candidate build tools, or only use them? The deployed systems are not Excel macros — they are applications with their own architecture, user manuals, training workflows, and production deployment on real \$400M+ projects.

Visual CRP (Docs 501–502) — Document Control System

Doc 501	Visual CRP presentation — PPSX	System overview: replaces Citadon, Construmanager, Procore in field environments without internet connectivity.
Doc 502	Visual CRP User Training — Guayubín PPG	1,183 active documents · 9 engineering firms · 3 commands for 100% daily usage · tripartite flow Firm→Consortium→Client · 6 dynamic report types · 19-year competitive analysis.
502A	Visual CRP Methodology — The Quiet Document Office	PDF white paper explaining the architectural decision: why attribute-based filtering eliminates the 1,600-folder bottleneck. Theoretical foundation for the system's design philosophy.
502B	Infographic 01 — Breaking the Folder Barrier	PNG. Comparison: traditional folder systems vs. Visual CRP attribute-based navigation. 1,600+ folder architectures vs. dynamic text filters. 15,000+ records · 4,000+ documents shown at operational scale.
502C	Infographic 02 — Solving the Document Control Bottleneck	PNG. Three failure modes of traditional systems: 1,600-folder headache · connectivity gap · loss of field trust. Visual CRP advantage: dynamic keyboard commands · latest-revision access · 15,000+ records managed.
502D	Infographic 03 — Beyond the Folder	PNG. 'Folder Fatigue' vs. Dynamic Filtering. Comparison table: Navigation · Office Traffic · User Autonomy. The Quiet Office Result: field engineers verify approved revisions in seconds without leaving the construction site.

Proof of product maturity: Montegrande operates its irrigation canal phase with Visual CRP running autonomously today — without external support from the author. That is not a system dependent on its creator. That is infrastructure.

AC&P System (Docs 503–509) — Parametric Cost Estimation

Two-layer architecture: Layer 1 (BID budget) uses hidden variation factors — labor retentions (FICA, Medicare, workers comp, benefits) as hidden factor toward strategic bid price. Layer 2 (individual APU) uses real field factors: productivity rates, adjustment factors, waste factors.

Doc 503A	AC&P US — Greensboro NC · Active May 2026	99-page output · 6 functional sheets · bilingual EN/ES toggle. Grand Total: \$508,500. CSI MasterFormat Div. 26. Journeyman Electrician \$150/HR. Graybar/Rexel NC 2025-2026 prices.
503A xlsx	AC&P US — Excel Workbook (Live File)	Active Excel .xlsx file with 6 functional sheets · bilingual EN/ES toggle · inverse factoring engine · CSI MasterFormat Div. 26 · NC/NJ labor rates · Graybar/Rexel 2025-2026 pricing. Downloadable from https://visualcrp.com/AC&P.html
Docs 504– 509	AC&P Palomino Accommodations — 6 iterations	5 typologies · 34,297 items quantified · RD\$13,420,384 total. 6 budget iterations without rebuilding the model = real parametric system, not a manual spreadsheet.

Inverse factoring engine (unique feature): Calculates backward from target price to required factor. Dialog: 'Current Direct Costs = 433,415.37. Define desired: [450,000] → Factor = 1.10109337926671.' One number updates all unit prices simultaneously. This does not exist in HeavyBid, WinEst, or Sage Estimating — commercial systems only work direct (inputs → price). AC&P works both ways.

SALT System (Doc 510) — Survey Adjustment Rev. 4.0

Doc 510	SALT user manual + Method PB	Compass Rule · Crandall Rule · Transit Rule + Method PB (own methodology for removable bases in tunnel). AutoCAD API/DDE/COM integration. Video: youtube.com/watch?v=opBY0S1D1R0 — 50 pts · 32,390m · 1:516,577.
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ETL Links — Serie 517A — Universal ETL Motor

Serie 517A	Vinculaciones — Rev 5.0	ExecuteExcel4Macro: reads N source files WITHOUT opening them · zero corruption risk · 10× faster than Workbooks.Open. Field map in MENU sheet — zero hardcoding: 14 configured fields shown in this file (Fecha Informe · Código · Frente · Pared · Progresiva · Elevación · Nº Barras · Ø barra · Tipo Perno · Long. Perforación · Ø Perforación · Fecha Inyección · Relación A/C · Cantidad Cemento). Production output: 750+ Dywidag rock bolt records · Casa de Máquinas Palomino · Jan–May 2010 · source files C:\PR\IP(1).xls → P(750+).xls — each read without opening. Rev 5.0 · configurable to any field layout.
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SERIES 600–615

Engineering Reports & Case Studies

Change Order Analysis · Geodetic Control · Visual CRP External Validation · Monorail SNIP 17248 Infrastructure Audit

+36%	1:52,753	1,183	8
Concrete from +18% area (quadratic)	SALT factor precision	Guayubín docs certified	CERCONS validation docs

Doc 602	Tunnel over-excavation forecast — change order	+18% excess in D&B area → +18% shotcrete BUT +36% structural concrete because annular volume grows with the square of the radius. This is what justifies renegotiation. Most engineers register the area and stop.
Doc 603	TBM advance engineering report	Daily advance tracking with geotechnical correlation. KPIs and deviation from plan.
Doc 604	TBM superficial transverse survey	1:52,753 factor documented — 3× more precise than standard surveying.
Docs 605–607	Project control points — 3 projects	EGM2008 geoid documented across Las Placetas, Cibao-Sur, Bávaro-Miches. Systematic geodetic rigor across all assignments.
Docs 608–615	Visual CRP external validation — Guayubín	CERCONS/EGEHID Guayubín 2023: 1,183 documents, 9 engineering firms. Independent deployment. Validates that the system is deployable by any qualified user — not dependent on the author.
Doc 616	Monorail SNIP 17248 — Infrastructure Audit (Interactive Case Study)	Interactive case study based on a specialized independent audit. Executive synthesis, technical visualization and multi-format documentation ecosystem. Primary access: visualcrp.com/AUDIT.html — Dashboard interactivo v2.0. USD 932.8M public infrastructure decision (FITRAM-CCC-LP-2025-0001). 5 Notas Técnicas · 17 hallazgos.
Doc 616A	Executive Infographic — 01A	“El Dilema Ferroviario de Santo Domingo” — 17 hallazgos en 4 cuadrantes. USD 932.8M vs USD 3.3B alternative. tinyurl.com/2m83z4ma
Doc 616B	Executive Presentation — 02A	“The 3.3 Billion Infrastructure Audit” — 16 slides, bilingual EN/ES. Technical-financial synthesis for executive audience. tinyurl.com/mw92hat4
Doc 616C	Executive Podcast — 03A	Audio synthesis · 34 min · Deep Dive format with two speakers. Based on the complete NT I–V technical corpus. Synthesis: Ing. Piter J. Bonilla D. tinyurl.com/anpxackc
Doc 616D	Ex Ante Risk Report — 8 Jan 2026	Pre-bid technical-financial risk assessment · 13-slide executive report. Independent assessment completed before contract award, identifying operational, capacity and lifecycle cost risks through quantitative engineering analysis. Subsequent independent audit reached consistent conclusions and expanded the technical evidence.
Doc 616E	Independent Technical Video Series — Nov 2025–Jan 2026	Independent pre-award technical analysis · Six-part engineering series covering technology selection, operational benchmarking, capacity modelling and lifecycle cost evaluation. Produced before the independent audit, which later confirmed and expanded the principal technical findings through a comprehensive documentary review. youtube.com/@peterbonilla8239

SERIES 700-707

Quantity Take-off — Pay Application

Montegrande Dam | CDEEE/EGEHID Certified | \$500M+ Impasse Resolved

\$708K	928K m3	1.8M m3	5
Certified pay app Doc 702	Cut — PMG earthwork	Fill — PMG earthwork	Reports resolving impasse

Doc 701	Las Placetas boreholes — geotechnical QTO	Drilling services quantification with geological log correlation.
Doc 702	Palomino Phase 1 roadway — certified pay app	\$708,000 certified by CDEEE/EGEHID. Unit prices, quantities, and approval signatures. Not a calculation — a certified payment document.
Docs 703-707	PMG dam earthwork — 5 revision reports	Two-surface methodology: existing terrain TIN vs. design TIN. 928,682m3 cut + 1,797,261m3 fill. Five consecutive reports resolving a \$500M+ cubication impasse.

Methodology: 2-surface TIN comparison at 350 cross-sections every 5m. The impasse was a methodology dispute, not a measurement dispute. Establishing the correct reference surfaces and producing five auditable reports aligned all parties. This is change & risk management at the \$500M+ scale.

SERIES 800–807

Estimations & Drafting Service

Haina Project | Trasvase Hatillo | CAASD | ITRF-2000

4	228.20 Mm3	7	2
CAASD drawings in 48h	Haina reservoir capacity	Coding fields Doc 801	Projects — consulting

Doc 801	Document coding criteria — 7-field system	Proprietary document classification: Project + Discipline + Type + Subtype + Sequential + Rev + Status. Demonstrates systematic documentation methodology in independent consulting context.
Docs 802– 805	Haina Project — 4 drawings	4 technical drawings delivered in 48 hours for CAASD (Santo Domingo water authority). 228.20 Mm3 reservoir. Dual signature P. BONILLA Elaborador+Verificador. Demonstrates production speed.
Docs 806– 807	Trasvase Hatillo — multispectral DEM	Cordillera Central multispectral DEM processing. ITRF-2000 reference frame. Hydrological transfer study for water resource infrastructure.
807A	Reservoir Storage Analysis — Elevation-Volume & Elevation-Surface Curves	Embalse Guaigüí · Topografía Referencial · 2 pages PDF. Elevation/Volume curve: Elev.320.00=35.54 Mm3 · Elev.326.00=50.16 Mm3 · Elev.330.00=61.60 Mm3. Elevation/Surface curve: Elev.320.00=2.17 km2 · Elev.326.00=2.70 km2 · Elev.330.00=3.02 km2. WGS-84 georeferenced. Spillway & intake design support.
807B	Reservoir Storage Analysis — Useful Storage Tables (55 Scenarios)	Embalse Guaigüí · Estimación Volumen Útil · 11 pages PDF. 55 combinations of minimum operating level vs. spillway crest elevation. Columns 1–55 · Vol.Util (Mm3) range 0.78–54.10 Mm3 · Elevation range 300–330m. Direct input for spillway sizing, intake elevation optimization, and drought yield analysis.

SERIES 901-916

Excavation Advance Over Planning

Montegrande Dam | 16 Drone Periods | Early Warning System

3.74M	16	38%	69%
m3 total tracked	Drone survey periods	Sector C advance	Sector C deficit share

Sixteen consecutive drone surveys tracking excavation progress against plan. The Sector C early warning is the most transferable methodology in the entire portfolio for change & risk management roles.

Docs 901-906	Period reports — snapshots S-61 to S-70	Six reporting snapshots. Each report: actual vs. plan per sector, cumulative deficit/surplus, projection to completion.
Docs 907-916	Drone imagery — 10 aerial images	Direct field evidence from drone surveys. Visual confirmation of each measurement period.

Early warning demonstrated: Snapshot 70 identified that Sector C was at 38% of planned advance while holding 69% of total accumulated deficit. That is early warning, not crisis response. The client's engineers reallocated equipment before the delay became unrecoverable. This is progress monitoring with predictive value — not just reporting what happened.

CONTACT & ACCESS

Portfolio & Professional Information

This portfolio documents 25+ years of technical work across large-scale EPC infrastructure projects. All documents are indexed and accessible online. For inquiries, collaboration, or project-specific discussions, use the contact information below.

Piter J. Bonilla D. <i>Civil Engineer Office Engineer / Construction Administration EPC Infrastructure</i> EPC Infrastructure & Power Generation CODIA #17378 — República Dominicana Elizabeth NJ — United States Green Card No sponsorship required +1 (336) 268-7426 peterbonilla@hotmail.com	Online Portfolio Competency Evidence Index https://visualcrp.com/PortfolioCompetencyEvidenceIndex.html Portfolio Narrative https://visualcrp.com/PortfolioNarrative.html Portfolio — 146 indexed documents https://visualcrp.com/portfolio.html Resume online https://visualcrp.com/resume.html AC&P Estimating System + demo https://visualcrp.com/AC&P.html Visual CRP Document Control System https://visualcrp.com LinkedIn Profile https://www.linkedin.com/in/piter-bonilla-16244473
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Value proposition: When commercial software doesn't solve the problem, the solution is built and deployed in production. Five proprietary systems — AC&P (2004), SALT (2004), Visual CRP (2007), ETL Links (2008), ICOD (2020) — demonstrate 22 years of continuous iteration on the same technical architecture, deployed on projects ranging from \$300M to \$500M+. Not legacy systems — active infrastructure.